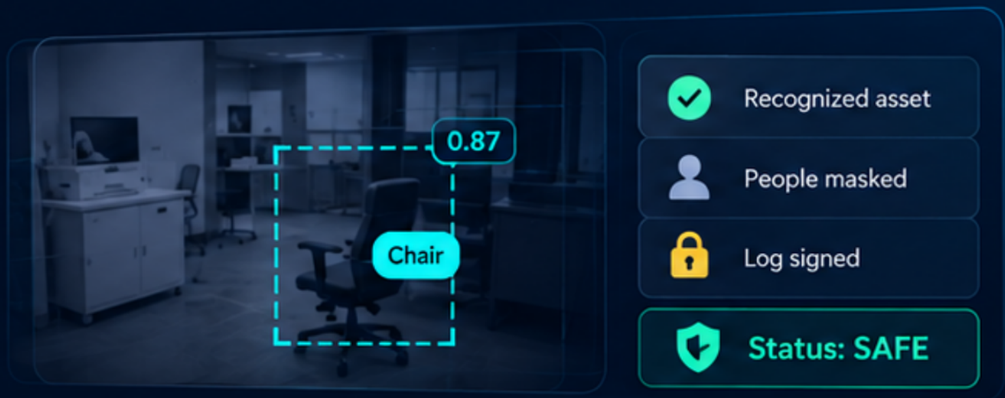
PROJECT INTELLIWATCH | L&T FINANCE x TECH TATVA

TRACE

Trusted Real-time Asset Confidence Engine

A software-only, privacy-first system that tells a branch not just what changed, but how sure it is about every asset, every second.



THE PROBLEM

Identify registered assets in a branch-like space, detect when their location or state changes, and raise an actionable alert backed by digital evidence.

OUR ANSWER

A software-only, privacy-first system that tells a branch not just what changed, but how sure it is about every asset, every second.

WHERE TODAY'S ASSET MONITORING FAILS

1 Blind alerts

Camera blocked or network down? Most systems stay silent or cry wolf. TRACE reports UNKNOWN and never guesses.

2 Privacy risk

Branch video exposes staff and customers. TRACE stores asset crops only; people are masked on-device.

3 Weak evidence

Editable logs fail audits. Every event is signed and hash-chained, with the head anchored off-box.

4 Alert fatigue

Laptops and chairs move all day. A policy layer (fixed vs. mobile, after-hours) keeps alerts rare and real.

THE DEV NATION: Dhyey Patel | Suchet Amaljari | Neha Rao | AIML Batch of 2027 | MIT Manipal

Solution & System Architecture

Software-only edge AI • Runs on laptop/low-cost box • No raw video leaves the branch

INPUT

Recorded branch clips
+ scripted events

- Move
- Cover lens
- Cut network
- Occlude

Live webcam / RTSP
(same path)

EDGE NODE Laptop / Pi 5 • Offline-capable

1 REGISTER
ROI, ID, policy,
3+ photos

2 SENSE
Health check, person mask,
ROI match

3 BELIEVE
State machine,
decaying confidence

4 PROVE
Masked crops,
signed hash chain

SQLite outbox • Signed events queue (offline)

ALERTS + DASHBOARD

Printer-01	SAFE
Safe-02	MOVED
Laptop-03	UNKNOWN
Cam-1	OBSTRUCTED

Incident #12 • MOVED • conf 0.93

Before | After | Verify ledger

10 BASELINE REQUIREMENTS COVERED

Register 1–2 | Detect 3 | Movement 4 | Missing 5 | Unexpected 6
Classes 7 | Evidence 8 | Alerts 9 | Robustness 10

OPEN-SOURCE STACK

OpenCV | ONNX Runtime | YOLO11n / OWLv2 | ByteTrack | MediaPipe
PyNaCl | FastAPI | SQLite | Streamlit | Docker

MOCK ASSETS

Printer, safe (cardboard), laptop, camera model, chair, stationery

Synthetic environment only • No real customer data

Engineering Mechanics

Four independent signals feed one confidence model

Signal 1: Location drift

Homography maps the box to the floor plan. Alert when drift exceeds a per-class threshold for N frames. Wall landmarks separate ‘camera moved’ from ‘asset moved’.

Signal 2: Identity drift

Embedding similarity to the Visual DNA below the asset's own threshold means same place, different object. A QR/AprilTag sticker separates identical models.

Belief model: the Doubt Engine

$$C(t) = C0 \cdot \exp(-\lambda \cdot t_unseen)$$

A verified sighting resets C to 1. Below 0.5 the state is UNKNOWN, never a false MOVED. λ is per class; calibration and hysteresis stop flapping.

Signal 3: State change

Small classifier trained on a few mock images per state: safe door open or closed, printer cover off, cable unplugged. Covers the ‘state-change’ clause.

Signal 4: Presence

MISSING needs proof: ROI visible, camera healthy, ROI matches the empty background. NEW_OBJECT is a persistent non-person blob matching no registered asset.

Tamper-evident ledger

$$h(n) = \mathrm{SHA256}(h(n-1) + \mathrm{crop} + \mathrm{event} + \mathrm{ts})$$

Ed25519-signed. Head hash anchored off-box (RFC 3161 or Telegram), so a full-chain rewrite is detected.

Anomaly classes (requirement 7)

ALERTS: MOVED · SWAPPED · STATE_CHANGED · MISSING · NEW_OBJECT · TAMPERED · UNKNOWN
System: CAMERA_OBSTRUCTED · SYSTEM_OFFLINE
Quiet: SAFE · OCCLUDED · RETURNED

Privacy by design

People are segmented in memory (an occlusion signal) and masked before anything is stored. No face recognition.

Policy layer

Fixed vs. mobile assets, after-hours mode, authorized-move acknowledgement, dedupe, cooldown, severity by criticality.

Validation Plan and Live Edge-Case Handling

Built for the Round 2 twist: obstruction, network loss, partial occlusion

Edge case	How TRACE responds
1 Camera obstructed	Frame-health check (blur, brightness, edge density, landmarks) raises CAMERA_OBSTRUCTED in about 2 s. Assets decay to UNKNOWN, MISSING is suppressed, staff alerted.
2 Network loss	Edge keeps alerting locally. Signed events queue sync on reconnect with a 'delayed' badge, and the chain is re-verified.
3 Occlusion / person in front	People mask plus visibility ratio and multi-frame voting give OCCLUDED. Low visibility lowers confidence; it never triggers an alert.
4 Lighting change	CLAHE, embedding normalisation and multi-lighting baselines keep SAFE stable.
5 Look-alike swap	Per-asset threshold plus optional tag gives SWAPPED, even for identical models.
6 Camera bumped / offline	Landmark drift gives TAMPERED and auto-recalibration. Heartbeat loss gives SYSTEM_OFFLINE.

TEST PROTOCOL & TARGETS

6 assets × 6 events | **20 trials** (= 720 events) | **Target** ≥ 99% precision

Targets will be replaced by measured values after testing:

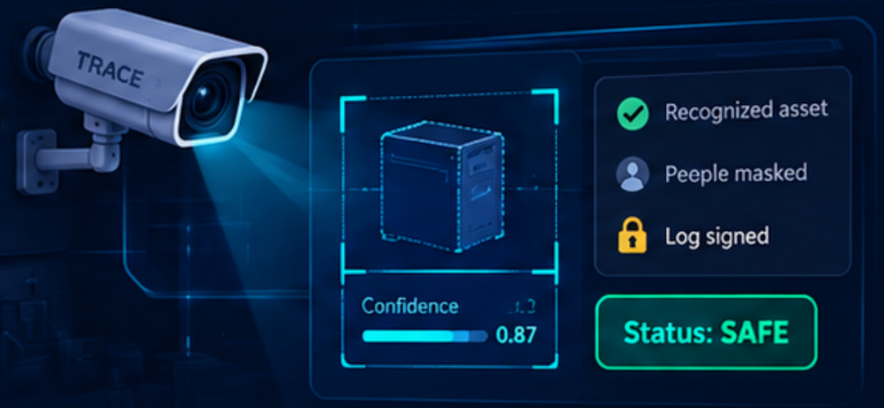
- **Precision** 99%+ on auto-confirmed alerts
- **Recall** 97%+ on MOVED, MISSING, SWAPPED
- **Under 1** false alarm per camera per day
- **MOVED** latency under 2 s (MISSING waits by design)
- Ledger tamper test: 100% detected, incl. full-chain rewrite
- **Report** a confusion matrix, P50/P95 latency and Wilson 95% intervals. Uncertain cases go to a review queue, not a blind alert.

Round 2 live demo: move | swap | cover the lens | unplug wifi | person walks through | ledger verified on screen

Every edge case maps to a visible dashboard state.

Why TRACE Wins and How It Scales

More accurate. More private. More scalable.



Aspect	Typical Solution	TRACE
Detection approach	Object detector + motion alert	Per-asset belief state with decaying confidence
Handles blocked view	Fails silently when blocked	Detects blindness and reports UNKNOWN honestly
Video storage	Stores raw video	Masks people on-device; keeps asset crops only
Log integrity	Editable log file	Signed, hash-chained ledger anchored off-box
New asset onboarding	Retrain per new asset	3-photo registration, no training
Alert behaviour	Alerts on every move	Policy layer: fixed vs. mobile, after-hours, authorized move

Feasibility and Cost

- One laptop + webcam
- Edge box (2 IP cameras + UPS)
- ₹25–40k per branch
- Open-source zero licence cost

Prototype vs Production

Prototype (Proof of Concept)
One Python process, SQLite, FastAPI, Streamlit.

Production (Scalable)
PostgreSQL, NIM, MQTT, Keycloak, Prometheus, AGPL YOLO (Apache-2.0) or RT-DETR.

Threat Model

- Camera tampering
- Photo/screen spoofing
- Insider log deletion
- Edge compromise

Mitigated by signed evidence, off-box anchor and multi-channel alerts.

Scale and Roadmap

1. **Pilot** — Two-camera fusion, ESP32/RFID on safes
2. **Rollout** — Fleet dashboard across branches
3. **Expansion** — Same pipeline for ATMs, vaults, gold counters

TRACE does not just watch assets. It proves, with evidence and honest confidence, what it knows.